

Sai Nitish Reddy Kapa

Forward Deployed Engineer | AI Governance

Email: kapasainitishreddy@gmail.com | Location: United States | LinkedIn: [linkedin.com/in/kapasainitishreddy](https://www.linkedin.com/in/kapasainitishreddy) | Portfolio: kapasainitishreddy.github.io/portfolio

SUMMARY

Forward Deployed Engineer who embeds with teams to design, integrate, and deploy AI systems in live customer environments. FDE-certified (Educative) with hands-on experience building multi-agent systems, RAG pipelines, text-to-SQL agents, and AI governance tooling — including NIST AI RMF-aligned risk registries and LLM evaluation/audit platforms. Currently completing an AI Governance certification. Background spans data analytics, AI model evaluation, and cross-functional delivery.

SKILLS

AI & Agent Engineering: LangGraph, CrewAI, LLM Orchestration, Retrieval-Augmented Generation (RAG), Prompt Engineering, Model Context Protocol (MCP), Text-to-SQL, LLM Evaluation, AI Governance (NIST AI RMF, EU AI Act)

Data & Integration: Python, SQL, PostgreSQL, Supabase, FastAPI, API Integration, Data Pipelines, Power BI

Deployment & Delivery: Next.js, TypeScript, React, CI/CD, Vercel, Netlify, GitHub Actions, Production Debugging, Monitoring

Platforms & Tools: OpenAI API, Anthropic Claude API, Vector Search / Embeddings, Git

EXPERIENCE

Forward Deployed Engineer (Freelance)

2024 - Present

Seed-Stage AI Startup (client and customer confidential under NDA)

- Embedded with a small founding team to take an AI product from prototype to production inside real customer environments
- Built and integrated LLM-powered workflows against the customer's live data, authentication, and internal tools, replacing brittle manual processes
- Instrumented evaluations, logging, and human-in-the-loop review so deployed systems stayed accurate, safe, and auditable
- Shipped an AI support copilot that cut first-response time approximately 60% on covered ticket types and resolved approximately 45% of routine tickets with copilot-drafted answers, live in under 4 weeks
- Built and automated a production data and reporting pipeline that eliminated approximately 20 analyst-hours per week and achieved 99.5% uptime across the engagement
- Designed a human-in-the-loop customer onboarding agent that cut onboarding time from approximately 3 weeks to 4 days, with zero irreversible actions taken without human approval

Business Data Analyst

Aug 2024 - Present

Augmentare Inc.

- Deploy internal data tooling, dashboards, and reporting pipelines used directly by decision-makers
- Automated recurring reporting workflows, improving reporting efficiency by 28%
- Integrate data across business and technical systems and translate findings into executive-ready reports and presentations

AI Evaluation and Governance Specialist (Freelance)

Outlier AI

- Reviewed 1,800+ model responses for factual accuracy, safety, reasoning quality, and instruction following
- Documented 30+ recurring failure modes and translated them into structured, actionable feedback for engineering teams
- Supported human-in-the-loop review workflows that keep AI systems accountable and audit-ready

Data Analyst

Jun 2020 - Jan 2023

VN Technologies

- Built recurring dashboards and data pipelines that reduced analysis turnaround by 30%
- Partnered with cross-functional stakeholders to gather requirements, resolve inconsistencies, and manage revisions
- Produced client-ready reports, process documentation, and milestone updates for distributed teams

PROJECTS

Live project showcase: kapasainitishreddy.github.io/newresume — Data analysis case studies: kapasainitishreddy.github.io/resumeanalytics

RiskLedger — AI Use Case Registry and Risk Tiering

Next.js, TypeScript, PostgreSQL, NIST AI RMF mapping

- Pre-deployment AI governance registry that catalogs AI use cases and tiers them by risk against NIST AI RMF and EU AI Act categories

- Built approval workflow and automated review-cadence reminders so systems are reviewed before, not after, launch

Research Swarm — Multi-Agent Enterprise Research System

LangGraph, CrewAI, Python, FastAPI

- Role-based agent crew (planner, researcher, verifier, writer) that decomposes research questions, calls tools independently, and cross-checks claims before synthesis
- Implemented full tracing of every tool call and agent decision for auditability

QueryPilot — Text-to-SQL Data Analyst Agent

Python, LangChain, SQL, PostgreSQL, FastAPI

- Converts plain-English business questions into validated SQL, executed against a read-only sandbox before running
- Returns the exact query alongside every answer for auditability, with automatic chart generation for numeric results

MCP Bridge — Enterprise Tool Connector for AI Agents

TypeScript, Model Context Protocol (MCP), Node.js, PostgreSQL

- Model Context Protocol server exposing internal tools (ticketing, CRM, internal APIs) to any MCP-compatible AI agent through one governed, permissioned interface
- Built per-agent permission scoping, rate limiting, and full call logging for audit and debugging

GovSeal — Deployment Governance Console

Next.js, TypeScript, PostgreSQL, LLM Evaluation, Monitoring

- Governance console layered on deployed AI systems: continuous evaluation runs, drift monitoring, human-approval workflow, and a tamper-evident audit log

TraceGrid — LLM Evaluation and Audit Platform

Python, Next.js, TypeScript, Vector Search

- Evaluation workbench that captures the full trace of every LLM response and scores it for factuality, safety, bias, and policy compliance
- Regression grid surfaces quality drops release-over-release; failing cases route to a red-team review queue

CERTIFICATIONS

- Forward Deployed Engineer — Educative (Certified)
- AI Governance — In Progress (Responsible AI and AI Governance)
- Google IT Support Professional Certificate — Google

EDUCATION

M.S., Data Science

University at Buffalo

B.Tech, Electronics and Communication Engineering

IIIT Trichy (Indian Institute of Information Technology, Tiruchirappalli)